

1. At time t seconds, $t > 0$, a particle P is at the point with position vector $\mathbf{r}$ m, where

$$\mathbf{r} = (t^4 - 8t^2)\mathbf{i} + \left(6t^2 - 2t^{\frac{3}{2}}\right)\mathbf{j}$$

- (a) Find the velocity of P when P is moving in a direction parallel to the vector $\mathbf{j}$ (4)

- (b) Find the acceleration of P when $t = 4$ (3)
-

2.

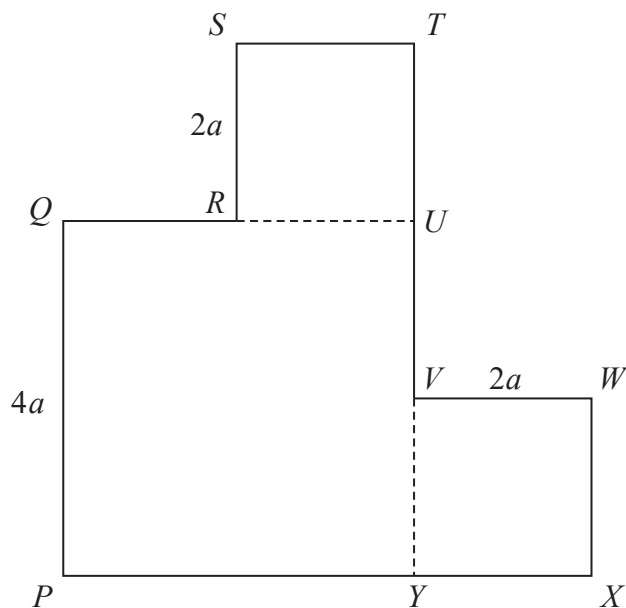


Figure 1

Figure 1 shows a template where

- $PQUY$ is a uniform square lamina with sides of length $4a$
- $RSTU$ is a uniform square lamina with sides of length $2a$
- $VWXY$ is a uniform square lamina with sides of length $2a$
- the three squares all lie in the same plane
- the mass per unit area of $VWXY$ is **double** the mass per unit area of $PQUY$
- the mass per unit area of $RSTU$ is **double** the mass per unit area of $PQUY$
- the distance of the centre of mass of the template from PX is d

(a) Show that $d = \frac{5}{2}a$

(5)

The template is freely pivoted about Q and hangs in equilibrium with PQ at an angle of θ to the downward vertical.

(b) Find the value of $\tan \theta$

(6)

The mass of the template is M

The template is still freely pivoted about Q , but it is now held in equilibrium, with PQ vertical, by a horizontal force of magnitude F which acts on the template at X . The line of action of the force lies in the same plane as the template.

(c) Find F in terms of M and g

(3)

3.

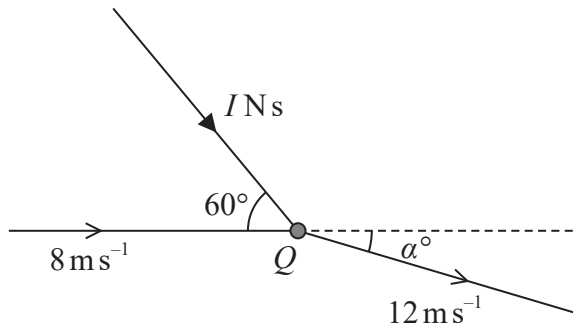


Figure 2

A particle Q of mass 0.25 kg is moving in a straight line on a smooth horizontal surface with speed 8 m s^{-1} when it receives an impulse of magnitude $I \text{ N s}$.

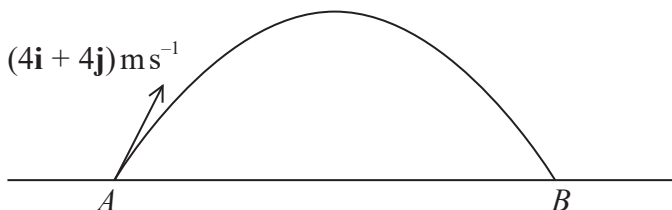
The impulse acts parallel to the horizontal surface and at 60° to the original direction of motion of Q .

Immediately after receiving the impulse, the speed of Q is 12 m s^{-1}

As a result of receiving the impulse, the direction of motion of Q is turned through α° , as shown in Figure 2.

Find the value of I

4.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are unit vectors, with $\mathbf{i}$ horizontal and $\mathbf{j}$ vertical.]**Figure 3**

The fixed points A and B lie on horizontal ground.

At time $t = 0$, a particle P is projected from A with velocity $(4\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$

Particle P moves freely under gravity and hits the ground at B , as shown in Figure 3.

At time T_1 seconds, P is at its highest point above the ground.

(a) Find the value of T_1

(2)

At time $t = 0$, a particle Q is also projected from A but with velocity $(5\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$

Particle Q moves freely under gravity.

(b) Find the vertical distance between Q and P at time T_1 seconds, giving your answer to 2 significant figures.

(3)

At the instant when particle P reaches B , particle Q is moving at α° below the horizontal.

(c) Find the value of α .

(4)

At time T_2 seconds, the direction of motion of Q is perpendicular to the initial direction of motion of Q .

(d) Find the value of T_2

(3)

5. A cyclist is travelling on a straight horizontal road and working at a constant rate of 500 W.

The total mass of the cyclist and her cycle is 80 kg.

The total resistance to the motion of the cyclist is modelled as a constant force of magnitude 60 N.

- (a) Using this model, find the acceleration of the cyclist at the instant when her speed is 6 m s^{-1}

(4)

On the following day, the cyclist travels up a straight road from a point A to a point B .

The distance from A to B is 20 km.

Point A is 500 m above sea level and point B is 800 m above sea level.

The cyclist starts from rest at A .

At the instant she reaches B her speed is 8 m s^{-1}

The total resistance to the motion of the cyclist from non-gravitational forces is modelled as a constant force of magnitude 60 N.

- (b) Using this model, find the total work done by the cyclist in the journey from A to B .

(5)

Later on, the cyclist is travelling up a straight road which is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{20}$

The cyclist is now working at a constant rate of P watts and has a constant speed of 7 m s^{-1}

The total resistance to the motion of the cyclist from non-gravitational forces is again modelled as a constant force of magnitude 60 N.

- (c) Using this model, find the value of P

(4)

6.

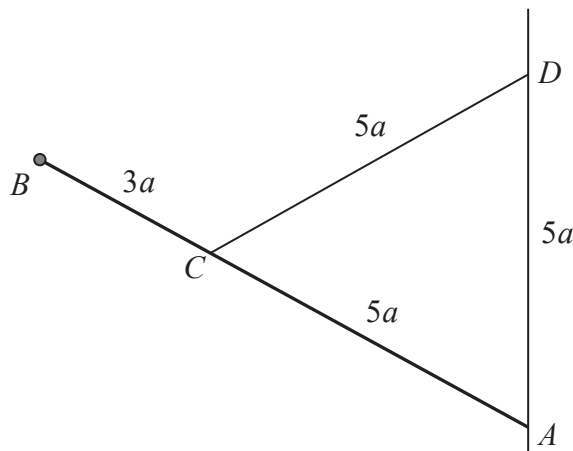


Figure 4

A uniform rod AB has length $8a$ and weight W .

The end A of the rod is freely hinged to a fixed point on a vertical wall.

A particle of weight $\frac{1}{4}W$ is attached to the rod at B .

A light inelastic string of length $5a$ has one end attached to the rod at the point C , where $AC = 5a$.

The other end of the string is attached to the wall at the point D , where D is above A and $AD = 5a$, as shown in Figure 4.

The rod rests in equilibrium.

The tension in the string is T .

(a) Show that $T = \frac{6}{5}W$ (3)

(b) Find, in terms of W , the magnitude of the force exerted on the rod by the hinge at A . (6)

7. Particle P has mass $4m$ and particle Q has mass $2m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface.

Particle P collides directly with particle Q .

Immediately **before** the collision, the speed of P is $2u$ and the speed of Q is $3u$.

Immediately **after** the collision, the speed of P is x and the speed of Q is y .

The direction of motion of each particle is reversed as a result of the collision.

The total kinetic energy of P and Q after the collision is half of the total kinetic energy of P and Q before the collision.

(a) Show that $y = \frac{8}{3}u$ (6)

The coefficient of restitution between P and Q is e .

(b) Find the value of e . (3)

After the collision, Q hits a smooth fixed vertical wall that is perpendicular to the direction of motion of Q .

Particle Q rebounds.

The coefficient of restitution between Q and the wall is f .

Given that there is no second collision between P and Q ,

(c) find the range of possible values of f . (3)

Given that $f = \frac{1}{4}$

(d) find, in terms of m and u , the magnitude of the impulse received by Q as a result of its impact with the wall. (2)